

Hexadecimal -- Guided Notes

Unit tp-1.3 | CompTIA Network+ N10-009 Obj. FC0-U71 1.2 | N10-008 FC0-U71 1.2

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

Hexadecimal is a base-_____ number system. It uses the digits 0 through 9 and the letters _____ through _____. The letter A represents the decimal value _____, and the letter F represents _____.

Question 2

Convert the hexadecimal value 2F to decimal. Show your work:

2 is in the 16^1 position = $2 \times$ _____ = _____

F (15) is in the 16^0 position = $15 \times$ _____ = _____

Total = _____

Question 3

To convert binary to hexadecimal, group binary digits into sets of _____ starting from the right.

Convert the binary number 11010110:

Left group: _____ in binary = _____ in hex

Right group: _____ in binary = _____ in hex

Result: _____

Question 4

The prefix _____ is used in programming to indicate a hexadecimal value (e.g., 0xFF). The hex value 0xFF equals _____ in decimal, which in binary is _____ -- the maximum value of one _____.

Question 5

Hex is used for _____ addresses, which are 48-bit hardware identifiers written as six pairs of hex digits (e.g., AA:BB:CC:DD:EE:FF). Each pair represents _____ bits. Hex is also used for _____ codes in design (e.g., #FF5733).

Question 6

Hex is also used for _____ addresses, where the location of data in RAM is expressed as a hex value. Programmers prefer hex over binary because it is more _____ while still mapping cleanly to binary -- every hex digit represents exactly _____ binary bits.

Question 7

Convert the decimal number 255 to hexadecimal. Show your work:

255 divided by 16 = _____ remainder _____

The remainder _____ is the rightmost hex digit = _____

The quotient _____ is the next digit

Result: _____

Question 8 -- Real World Application

REAL WORLD: A network technician reads a system log and sees a memory address listed as 0x1A4F. A new employee asks why the address is not just written in decimal. Explain why engineers use hexadecimal instead of decimal or binary for memory addresses, using at least two reasons.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/tp-1-3-hexadecimal/tp-1-3-hexadecimal-answer-key.pdf
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